

AI-ASSISTED CT CONTRAST BOLUS TRACKING

Major Project Phase II Report *submitted by*

Ms. Adithi Pai S
(NNM22IS008)

Ms. Anusha Sharma
(NNM22IS022)

Ms. Hemani Sunil Rao
(NNM22IS057)

Mr. Hitesh
(NNM22IS059)

To



(Estd. under Section 3, UGC Act 1956)

(Placed under Category 'A' by MHRD, Govt. of India; Accredited as 'A+' Grade University by NAAC)

In partial fulfilment of the requirements for the award of

BACHELOR OF TECHNOLOGY in INFORMATION SCIENCE AND ENGINEERING

DEPARTMENT OF INFORMATION SCIENCE AND ENGINEERING

NMAM INSTITUTE OF TECHNOLOGY, NITTE

May 2026

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Under the Guidance of

Dr. Preethi Salian K
Associate Professor
Department Of Information Science and Engineering
NMAM Institute of Technology

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Department of Information Science and Engineering**CERTIFICATE**

This is to certify that project report of the project work titled “**AI-Assisted CT Contrast Bolus Tracking**” is a bonafide research work carried out by **Ms. Adithi Pai S (NNM22IS008)**, **Ms. Anusha Sharma (NNM22IS022)**, **Ms. Hemani Sunil Rao (NNM22IS057)**, **Mr. Hitesh (NNM22IS059)** under the guidance of **Dr. Preethi Saliyan K**, Associate Professor, Department of Information Science and Engineering, NMAM Institute of Technology, Nitte and co-guidance of **Mr. Shashi Kumar Shetty**, Course Coordinator, Assistant Professor (Grade II), Department of Medical Imaging Technology, Nitte Institute of Allied Health Sciences.

This same is being submitted to the Nitte (Deemed to be University) in partial fulfilment of the requirements for the award of **Bachelor of Technology in Information Science and Engineering**.

Name & Signature of Guide(s)**Name & Signature of HOD****Head of the Institution****Project Viva voce****Name of the Examiners****Signature with Date**1. _____

2. _____

Alignment with Sustainable Development Goals

The project titled “**AI-ASSISTED CT CONTRAST BOLUS TRACKING**” supports the following
Sustainable Development Goals (SDGs):

SDG No	Goal Name	Description	Icon
3	Good Health and Well being	Ensure healthy lives and promote well-being for all at all ages	 The icon for SDG 3 is a green square. It features the number '3' in white at the top left, followed by the text 'GOOD HEALTH AND WELL-BEING' in white. Below the text is a white line graph with a heart shape at the end, symbolizing health and well-being.
9	Industry, Innovation and Infrastructure	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation	 The icon for SDG 9 is an orange square. It features the number '9' in white at the top left, followed by the text 'INDUSTRY, INNOVATION AND INFRASTRUCTURE' in white. Below the text is a white geometric design of three cubes stacked together, symbolizing industry and infrastructure.
12	Responsible Consumption and Production	Ensure sustainable consumption and production patterns	 The icon for SDG 12 is a brown square. It features the number '12' in white at the top left, followed by the text 'RESPONSIBLE CONSUMPTION AND PRODUCTION' in white. Below the text is a white infinity symbol, symbolizing sustainable consumption and production patterns.

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ABSTRACT

Contrast-enhanced computed tomography (CT) is a fundamental part of current diagnostic imaging, with the accuracy of such examinations fundamentally dependent on the precise arrival time of the contrast bolus at the target vessel. Conventional bolus tracking methods require a predetermined Hounsfield Unit (HU) threshold and operator-driven decisions, which result in inter-patient variability, poor image quality, and repeat acquisitions, all of which contribute to an increased radiation dose.

In this study, we introduce a new machine learning-based bolus tracking prediction system that predicts the ideal bolus tracking time before contrast injection using seven easily obtainable pre-scan clinical factors: patient age, sex, height, weight, contrast agent name and volume, and injection flow rate. We constructed a dataset of 1,108 patient records, which we enriched using domain-specific feature engineering: BMI and dose per kilogram. We trained and evaluated four regression models: Linear Regression, Random Forest, XGBoost, and a Multi-Layer Perceptron (MLP).

The Random Forest Regressor produced the best results, with a Mean Absolute Error (MAE) of 2.717 seconds on the held-out test set and a cross-validated MAE of 2.693 seconds (± 0.737 seconds) across five folds. Feature importance analysis revealed patient weight, BMI, and age as the top predictors, explaining over 54% of model importance, all in line with basic physiological principles governing contrast transit dynamics. The model was implemented as a web application using Streamlit, taking clinical inputs and producing estimated tracking time outputs along with uncertainty bounds of ± 2.693 seconds, forming a pre-scan decision-support tool for radiographers.

The predictions were done on four other clinical records that were not part of the training data set, where predictions within 3 seconds were successfully made for three out of the four cases. The one case where the prediction could not be made is that of a patient whose measured bolus tracking time was unusually high, standing at 26 seconds. This is a limitation of the model that was expected and that we wanted to address by collecting more data in this region of the distribution.

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NOMENCLATURE

Terms	Meaning
AI	Artificial Intelligence
ANN	Artificial Neural Network
BMI	Body Mass Index
CT	Computed Tomography
CTA	Computed Tomography Angiography
EDA	Exploratory Data Analysis
HU	Hounsfield Unit
MAE	Mean Absolute Error
MLP	Multi-Layer Perceptron
PACS	Picture Archiving and Communication System
ReLU	Rectified Linear Unit activation function
RFE	Recursive Feature Elimination
RIS	Radiology Information System
RMSE	Root Mean Square Error
ROI	Region of Interest
TAVI	Transcatheter Aortic Valve Implantation
XGBoost	Extreme Gradient Boosting machine learning algorithm

CHAPTER 1

INTRODUCTION

Contrast enhanced CT plays an important role in imaging because it facilitates visualization of blood vessels, internal organs, and diseases. However, the accuracy of images depends greatly on the correct timing of contrast enhancement. The slightest differences in the timing of scanning can have a great influence on the results obtained during the scan.

1.1 What is Bolus Tracking?

Contrast-enhanced computed tomography (CT) is one of the most widely used diagnostic imaging modalities in modern medicine. When a patient undergoes a contrast-enhanced CT scan, an iodine-based contrast agent is injected intravenously. As this agent travels through the bloodstream, it temporarily increases the X-ray attenuation of vascular structures and surrounding tissues, making blood vessels, organs, and lesions far more conspicuous on the resulting images.

Bolus tracking is the clinical technique used to determine the precise moment at which to initiate the diagnostic CT scan, that is, the moment at which the contrast agent has arrived at the target vessel or organ and enhancement is at its peak. The term "bolus" refers to the compact volume of contrast agent injected rapidly as a single dose. Tracking refers to the continuous, real-time monitoring of that bolus as it circulates through the body.

In current clinical practice, bolus tracking works as follows: after the contrast injection begins, repeated low-dose monitoring scans are acquired every one to two seconds at a fixed anatomical level most commonly over the descending aorta. Each monitoring scan measures the CT attenuation in Hounsfield Units (HU), within a user-defined region of interest (ROI). When this attenuation value rises above a predefined threshold typically 150 to 180 HU above the baseline, the system detects that the contrast agent has arrived, and the full diagnostic scan is triggered either automatically or by the radiographer.

While bolus tracking is now widely adopted in clinical workflows, it has evolved significantly over time to address earlier limitations in scan timing techniques.

1.2 History of Bolus Tracking

1.2.1 Manual Timing Era (1980s)

In the early years of contrast-enhanced CT, scan timing was determined entirely by fixed empirical delays following contrast injection. Radiographers applied a uniform post-injection delay typically ranging from 25 to 70 seconds regardless of individual patient physiology. There was no real-time monitoring of contrast arrival, and the approach was wholly operator-dependent, resulting in considerable variability in image quality across patients with differing cardiac output, body habitus, and vascular anatomy.

1.2.2 Test Bolus Technique (Early 1990s)

To solve the problems associated with fixed delays, a technique called test bolus was developed. This involved administering a small amount of contrast agent, acquiring low dose images in serial fashion at the target vessel to form a time density curve. From this curve, the peak enhancement time was calculated to determine the appropriate delay before giving the complete dose of contrast agent. Although much more accurate than the previous techniques, it increased time, contrast volume, and post processing efforts [10,12].

1.2.3 Automated Bolus Tracking (Late 1990s)

The technique of automated bolus tracking emerged as the prevailing method in the clinic. An ROI was set up around a target vessel, which was generally either the descending aorta or pulmonary artery, and a series of low dose monitoring images was performed at intervals of one to two seconds after the contrast injection. Once the ROI surpassed a certain level of Hounsfield units (HU), usually between 150-180 HU, the scan was automatically triggered after a post-trigger delay. This way, there was no need for a test bolus injection, and the scan was timed based on patient characteristics [3].

Even though automated bolus tracking greatly enhances the accuracy of scan timing, it is still dependent on a certain threshold and decisions made by the operator. Therefore, the current project aims to use an artificial intelligence-based approach to improve the consistency of scan timing using patient-specific parameters.

1.3 Need for AI in Healthcare

The use of Artificial Intelligence (AI) is one of the most recent developments in healthcare technology. As the volume of medical images grows, along with the need for quick and accurate diagnosis, traditional manual methods of processing data become inadequate. The use of AI makes it possible to work with large volumes of clinical and imaging data, detect hidden patterns, and assist in decision-making.

In the field of radiology, there is great potential in using AI to analyse images, detect diseases, optimize workflow, and plan scans automatically. Algorithms of machine learning make it possible to analyse specific physiological and imaging parameters of a particular patient and obtain information which is hardly achievable with the help of any traditional rule-based approaches. Thus, there is no need for the operator to make a decision and take account of all the possible variations.

When it comes to contrast-enhanced CT imaging, the traditional bolus tracking technique depends greatly on certain thresholds and decisions made by radiographers. There are many factors that affect the contrast transit time, such as cardiac output, body habitus, age, and vascular conditions, which makes the application of traditional protocols unreliable. The use of AI could help solve the problem, as it would learn from patient and scan history to predict the best scanning moment.

In addition to improving diagnostics, the use of AI will enhance patient safety, as it will lead to a decrease in the number of repeat examinations and radiation dose exposure. Thus, the application of AI in bolus tracking is very promising.

1.4 Conventional Clinical Workflow and Its Limitations

In the absence of AI tools, the standard workflow in clinical practice for contrast-enhanced CT scans includes the following steps:

1. A ROI is specified by the radiographer on the target vessel (the aorta, primarily) based on the pre-contrast localizer image.
2. The threshold HU value is selected manually according to either department protocols or radiographers' preferences rather than patients' conditions.
3. Contrast administration is performed, and then monitoring images are acquired after a specified delay period (usually 8-10 seconds) from contrast injection.
4. HU values are monitored in the ROI with a certain frequency (every 1-2 seconds)

until the threshold is achieved.

5. After reaching the required HU value, a specified post-trigger delay period (typically 5-8 seconds) is allowed for the patient to change position and prepare for breath-holding.

Obviously, the above-listed workflow has some limitations:

- Interoperator variability: The choice of threshold HU, ROI localization, and post-trigger delay are determined by the radiographer; therefore, there will be discrepancies.
- Physiological variability: The presence of heart dysfunction (poor pumping), obesity (higher volume of blood), and abnormal vasculature affects contrast arrival times. A predetermined threshold does not consider these parameters.
- Radiation exposure: Due to inappropriate timing, repeat scans can be required, leading to higher doses of ionizing radiation.

The above-mentioned limitations indicate a crucial flaw in the existing process.

1.5 Problem Statement

The process of bolus tracking manually is lengthy and prone to errors by the operator, hence causing a lot of inconsistency in scan timings. This inconsistency in scan timing affects the diagnostic accuracy since it causes poor contrast enhancement, which in turn results in the need to redo the scan, hence subjecting the patients to higher doses of radiation. These issues show the importance of developing an intelligent automated solution that will be able to predict the right scan timing using patient-specific data.

Considering the above challenges, this research project aims at achieving the following objectives:

1. Creating a clean and reliable dataset with varying patient physiological and protocol parameters.
2. Increasing the accuracy of diagnosis by ensuring optimal scan timing using data predictions.
3. Developing an automated bolus tracking system powered by AI that eliminates the need for manual scanning while minimizing the amount of radiation used.

CHAPTER 2

LITERATURE SURVEY

The timing of the contrast-enhanced CT imaging is a challenging issue, but it is crucial since the timing of the scan can affect the visibility of blood vessels. To address this problem, scientists have been working on ways to optimize the timing of the scan by varying the delay time before the scanning process, considering the influence of a person's body on the distribution of contrast in the bloodstream, and coming up with ways to predict the best timing for the scan. However, there are still some issues that should be addressed, and the next section will provide an overview of existing research in this field.

2.1 Review of Related Work

The research related to bolus tracking in CT has covered a wide range of topics, such as the impact of patient physiology on contrast timing, optimization of scan protocols, and automation of various aspects. C. T. Jensen et al. [4] found that delayed bolus triggering events are significantly associated with poor cardiac function, wherein the decreased cardiac output results in delayed peak aortic enhancement and suboptimal timing of portal venous phase imaging. Thus, the contrast transit time depends considerably on a patient's physiological condition, emphasizing the necessity of taking a patient-specific approach. In turn, Shashi Kumar et al. [2] identified the main parameters impacting patient-to-patient differences in enhancement timing, such as injection rate, contrast volume, body weight, and cardiac output.

Optimization of scan timing through post-trigger delays has been the focus of several studies. Specifically, J. Yu et al. [6] showed that patient-specific delays based on body weight and contrast volume result in superior arterial and portal venous enhancement compared to constant delays, although this method applies simple formulae that may fail to address complex cases. K. Qi et al. [17] and L. Nas et al. [18] applied this approach to aortic and lower extremity CT angiography, respectively, with an improvement in image quality observed in both cases. However, these methods require manual design of the formulae used, which makes them hard to generalize to other protocols. Besides, Y. Noda et al. [5] confirmed that the optimal delay is very sensitive to variables, such as contrast dose and injection rate.

Finally, some researchers have investigated the improvements of various stages of bolus tracking. For instance, A. T. Li et al. [8] suggested an automated technique for bolus tracking during abdominal CT, which improves the consistency of timing. Nevertheless, this algorithm is applicable to a limited dataset. Similarly, S. Watanabe et al. [13] proposed an automated measurement of Hounsfield Units to increase the reproducibility; however, they do not predict the trigger point. The improvements of bolus triggering were reported by M. D. Yuan et al. [12] in the case of coronary CT angiography, with a high level of triggering performance obtained; however, this technique is not readily generalizable. Moreover, E. Haneda et al. [20] offered a novel method to estimate the contrast dynamics using pulsed X-ray projection; nevertheless, it requires significant computation. Finally, H. Wu et al. [14] showed that ROI positioning may lead to variability among users, whereas I. Heukensfeldt Jansen et al. [7] studied the estimation of contrast kinetics without the need for a higher radiation dose.

The analysis of the above works demonstrates the following: even though there have been considerable advances in this area, one limitation still exists – the lack of a unified approach for integrating all available pre-scan patient data in order to accurately predict scan timing.

2.2 Summary of Reviewed Literature

Table 2.1 summarizes the key studies reviewed, including their methodology, principal advantages, and identified limitations, arranged in chronological order.

Table 2.1 Summary of reviewed literature on CT contrast bolus tracking optimization and automation

Ref.	Year	Authors	Methodology	Advantages	Disadvantages
[4]	2021	C. T. Jensen et al.	Retrospective analysis linking delayed trigger timing with cardiac function in abdominal CT.	Shows cardiac output affects trigger timing; supports patient-specific approaches.	No predictive model; limited to a specific cohort.

Ref.	Year	Authors	Methodology	Advantages	Disadvantages
[6]	2022	J. Yu et al.	Compared individualized vs fixed post-trigger delays using weight and contrast volume.	Improved enhancement; reduced variability.	Rule-based; limited to abdominal CT.
[8]	2023	A. T. Li, P. B. Noël, N. Shapira	Deep learning model for automatic trigger detection in bolus tracking.	Improves consistency; reduces manual effort.	Limited dataset; no prospective timing prediction.
[12]	2024	M. D. Yuan et al.	Compared AI-based triggering with conventional HU threshold method in coronary CTA.	Better consistency; improved enhancement.	Limited scope; not generalizable.
[13]	2024	S. Watanabe et al.	Deep learning for automated HU measurement in tracking images.	Improves reproducibility; reduces workload.	Does not predict timing; protocol-specific.
[14]	2024	H. Wu et al.	Studied impact of ROI	Identifies optimal	No predictive modelling; limited

Ref.	Year	Authors	Methodology	Advantages	Disadvantages
		al.	placement on image quality and trigger timing.	ROI placement.	generalizability.
[17]	2025	K. Qi et al.	Personalized delay formula using BMI and contrast flow rate.	Better enhancement uniformity.	Formula-based; limited to aortic CTA.
[18]	2025	L. Nas et al.	Compared individualized vs fixed delays in lower extremity CTA.	Improved image quality; fewer failed scans.	Manual calculation; no automation.
[20]	2025	E. Haneda et al.	Estimated contrast dynamics using pulsed X-ray projections and 5D model.	Reduces need for monitoring scans.	Computationally complex; not practical clinically.

2.3 Research Gap and Motivation

A review of the existing literature reveals that the current approaches to CT bolus tracking have several shortcomings such as relying on reactive systems that initiate scanning once the contrast agent has reached the ROI, using basic rule-based systems which are unable to account for more complex dependencies between patient and protocol variables, and lack of generalizability of the models to different machines and imaging protocols. Moreover, there is a lack of use of routinely collected pre-scan variables to develop predictive models. Finally, there are very few solutions that have been developed into deployable software. All of these gaps highlight the necessity to develop a predictive patient-specific model to predict bolus tracking time. Given the fact that contrast transit time varies significantly from individual to individual depending on their cardiac output and other bodily characteristics [2][4], and that machine learning models such as Random Forests [1] and XGBoost [3] can be successfully applied to handle structured clinical data, there is solid evidence to support the development of a predictive model. The improvements in image quality and reduction of non-diagnostic scans associated with better timing are an additional reason for the study [6][12][17], this study seeks to develop and implement a machine learning model for predicting the bolus tracking time using pre-scan parameters

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 Overview

In this chapter, the entire architecture of the proposed AI-Assisted CT Contrast Bolus Tracking Prediction System is going to be discussed. Inputs in the form of patient and scan parameters will be used to make predictions about the bolus tracking time. As per the architectural design of the system, each layer including inputs, processing, prediction, and outputs must be separate from one another.

Age, gender, height, weight, contrast volume, contrast type, and flow rate will be the inputs given to the system. Pre-processing of the inputs is done after which it will be fed into the trained Random Forest model to obtain predictions. Prediction results along with the uncertainty intervals will be returned back to the users.

The total number of layers in the system architecture is four. They include the Frontend Interaction Layer, Input Processing Layer, Inference Engine, and Output Layer. Artifacts used during the deployment of the model include the trained model, scaler, and feature column schema.

3.2 Architectural Layers

The system architecture has been designed in a way where all the processes are done sequentially in a single direction. Data passes through each layer after processing and is forwarded to the next layer. The following Fig. 3.1 shows the complete system architecture. This makes the system highly modular, easy to understand and modify in the future.

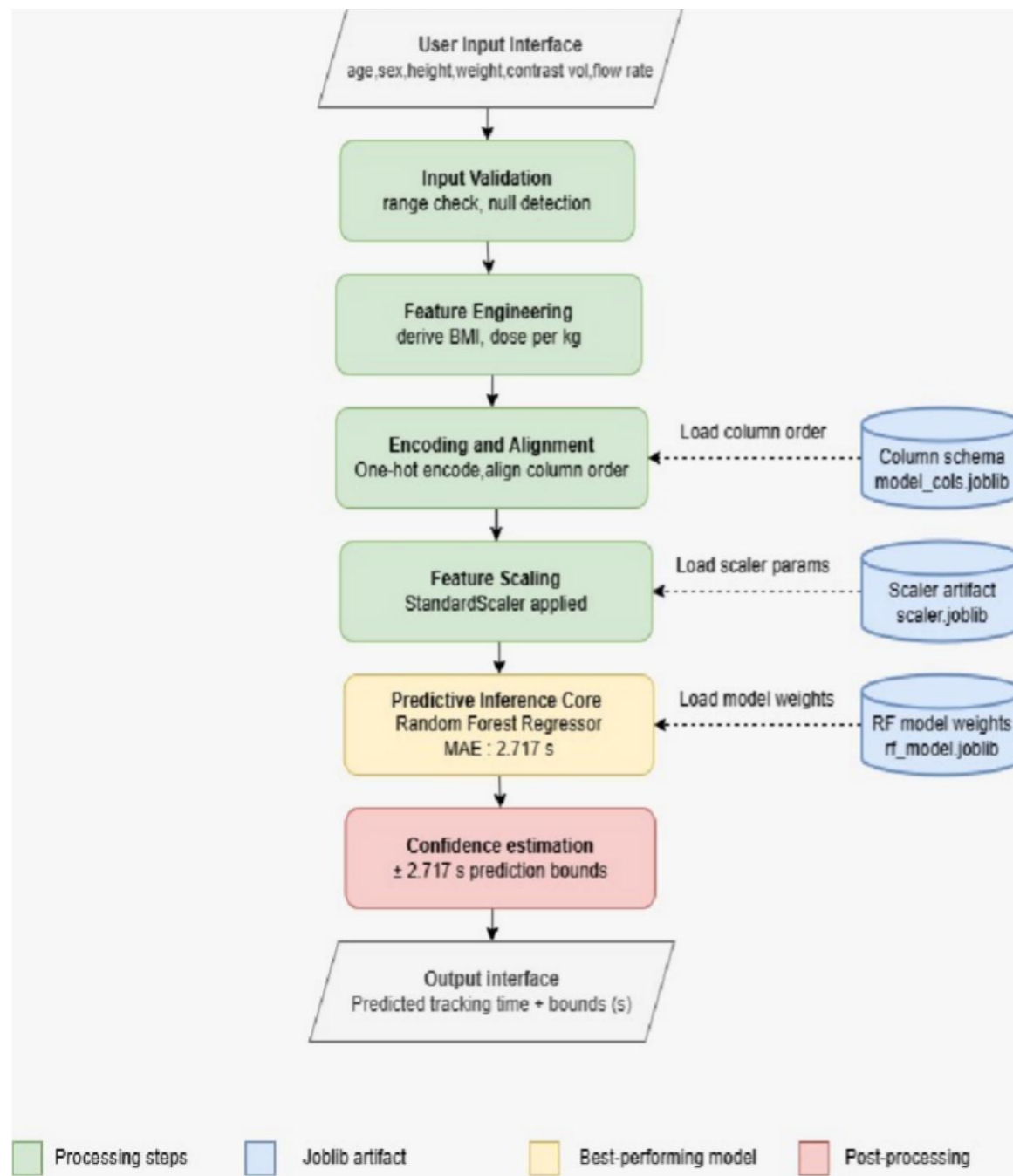


Fig. 3.1 System Architecture of the AI-Assisted Bolus Tracking

3.2.1 Frontend Interaction Layer

The frontend interaction layer represents the interface where interactions between clinicians such as radiographers and radiologists and the system occur. It is developed using Streamlit to allow for a light-weight web-based application without any installations.

The interface accepts the following seven clinical input parameters:

- Age (years)
- Height (cm)
- Weight (kg)
- Sex (Male/Female)
- Contrast Type (Iopromide / Iohexol)
- Contrast Volume (ml)
- Flow Rate (ml/s)

Inputs are taken using numeric boxes and dropdown menus. Simple validations (min/max ranges and null values) are performed on the inputs before passing them to the inference layer.

3.2.2 Input Processing Layer

The Input Processing layer converts clinical inputs into a feature vector for consumption by the trained model. It implements the same preprocessing pipeline used during training. The preprocessing pipeline comprises four consecutive steps:

1. Feature Engineering - Two more features are created:
 - BMI (Body Mass Index) based on height and weight
 - Dose per kg based on contrast volume and weight

This adds more precision to predictions by including physiological attributes specific to patients.

2. Categorical Encoding - One hot encoding of categorical features (Sex and Contrast Type) to convert them to numerical indicator variables. This allows the model to process them effectively.

3. Column Alignment - The new columns are aligned based on a schema specified in `model_cols.joblib`.
4. Feature Scaling - Standardization of all the columns using `StandardScaler` from `scaler.joblib`.

3.2.3 Inference Engine

The Inference Engine is the predictive module of the application. The model deployed here is a trained Random Forest Regressor stored as `rf_model.joblib`.

- It is an ensemble model with 100 decision trees
- Average output from all the trees is considered as a prediction
- Ensemble method makes the model more stable and less prone to variance. The model outputs the prediction of bolus tracking time (seconds)

To provide reliability, an uncertainty range is calculated using the model's Mean Absolute Error (MAE):

- Prediction uncertainty range: ± 2.717 seconds

3.2.4 Output Presentation Layer

The Output Layer displays the prediction results to the user via the Streamlit interface. The output includes:

- Predicted bolus tracking time
- Lower bound (Prediction – MAE)
- Upper bound (Prediction + MAE)

This provides the ability for the interpretation of results taking into consideration any variability that may exist. The system can be used as a decision support tool, allowing the radiographers to determine the scan time beforehand and avoiding the need for follow-up scans.

3.3 Serialized Model Artifacts

The deployment process is based on three serialized artifacts produced from the model training process. The artifacts are kept on disk and then loaded to memory when the application starts, which allows for inference to happen without retraining the model, and thus, at low latency. The artifacts along with the respective objects are shown in Table 3.1 below.

Table 3.1: Serialized Deployment Artifacts and Their Roles

Artifact File	Object Type	Role in Pipeline
final_rf_model.joblib	RandomForestRegressor	Trained ensemble model that generates the predicted bolus tracking time from the processed feature vector
scaler.joblib	StandardScaler	Normalizes input features using training-set statistics (zero mean, unit variance)
model_columns.joblib	List[str]	Stores the ordered feature column names to ensure correct alignment between input data and model expectations

All three artifacts are saved using Python’s Joblib library, which is optimized for efficiently storing and loading machine learning models containing large NumPy arrays, such as Random Forest ensembles. During application initialization, these artifacts are loaded once into memory and reused for all subsequent prediction requests. This design minimizes

runtime overhead by eliminating repeated disk access and ensures that inference latency is dominated primarily by the model's prediction step rather than data loading operations.

3.4 End-to-End Data Flow

The system follows a structured sequence of operations to transform raw clinical inputs into a predicted bolus tracking time. The complete inference process for a single patient record proceeds as follows:

1. **Raw Input Acquisition** - The radiographer enters seven patient-specific parameters through the Streamlit web interface.
2. **Input Validation** - The system performs range and type checks to ensure valid and physiologically plausible inputs.
3. **Feature Engineering** - Two additional features, BMI and Dose per Kilogram, are computed, expanding the feature set from seven to nine.
4. **Categorical Encoding** - Categorical variables (Sex and Contrast Type) are transformed into binary features using one-hot encoding.
5. **Column Alignment** - The feature vector is reordered according to the schema stored in `model_columns.joblib` to maintain consistency with training.
6. **Feature Scaling** - The aligned features are standardized using the pre-fitted `StandardScaler` loaded from `scaler.joblib`.
7. **Model Inference** - The processed feature vector is passed to the Random Forest model (`final_rf_model.joblib`), which generates a prediction by averaging outputs from 100 decision trees.
8. **Output Presentation** - The predicted bolus tracking time, along with a prediction uncertainty range of ± 2.717 seconds, is displayed to the clinician.

Table 3.2: End-to-End Inference Steps

No.	Step	Action
1.	Input	Radiographer enters 7 patient parameters via the Streamlit form
2.	Validate	Basic range and type checks on raw inputs
3.	Engineer	BMI and dose_per_kg computed; features expand to 9
4.	Encode	Sex and Contrast Type one-hot encoded
5.	Align	Columns matched with schema (model_columns.joblib)
6.	Scale	Features standardized using scaler.joblib
7.	Predict	Random Forest generates prediction (seconds)
8.	Output	Predicted time with uncertainty range of ± 2.717 s displayed

3.5 Deployment Environment

The system has been deployed as a standalone application written in Python, utilizing the Streamlit library to render the graphical interface within a web browser. This implementation does not depend on any cloud infrastructure, external databases, or network connectivity; thus it is suitable for use in local hospitals or even departmental workstations, including air-gapped clinical settings.

The application can be run from the command line using the Streamlit library, which will initialize a local HTTP server and launch the graphical interface within the default web browser. The serialized objects containing the trained models (final_rf_model.joblib, scaler.joblib, and model_columns.joblib) are stored locally and loaded only once at application startup.

Prediction queries are handled synchronously by a single Python process. Because both the model and its preprocessing pipeline have already been loaded into memory, the inference time remains under a second for each prediction query. The system was developed with data security in mind since no personal data is stored, logged, or transmitted externally during the usage of the system.

Table 3.3: Deployment Environment Dependencies

Library	Version	Purpose
Python	3.9+	Runtime environment
Streamlit	1.20+	Web application framework and UI
Scikit-learn	1.2+	Random Forest model and StandardScaler
Pandas	1.5+	Data processing and feature encoding
NumPy	1.23+	Numerical computations
Joblib	1.2+	Serialization of model artifacts

3.6 Key Architectural Design Decisions

Some of the decisions made regarding the design of the system include ensuring accuracy, efficiency, and clinical usability.

1. Inference Model – Random Forest – Of all the models considered including Linear Regression, Random Forest, XGBoost, and Artificial Neural Network (ANN), Random Forest Regressor performed the best with an MAE of 2.717 seconds when using cross-validation. This model was chosen because:
 - Strong generalization performance
 - Robustness to overfitting

- Stable predictions across different data splits
 - Interpretable feature importance
2. Retraining using the Full Dataset - The model was retrained using the full dataset consisting of 1,108 rows. This helps the model make use of all data to generalize better. The MAE obtained after cross-validation is an unbiased estimate of the error expected.
 3. Stateless and Synchronous Inference - The system processes each request independently without storing any patient data. This stateless design:
 - Ensures data privacy
 - Eliminates the need for databases
 - Simplifies deployment
 4. Fixed Uncertainty Calculation - Rather than using complex computational techniques, the system employs a fixed uncertainty range of ± 2.717 seconds based on cross-validation errors. The method is straightforward, effective, and easy to interpret clinically.

3.7 Advantages of the Proposed Architecture

There are multiple benefits of the proposed AI-assisted bolus tracking framework that make it applicable for clinical use and real-time decision support in contrast-enhanced CT imaging.

1. Modular Architecture - The framework consists of separate layers responsible for the input data processing, pre-processing, prediction, and output visualization. Such architectural design increases the maintainability and allows modifying any component separately from the whole system.
2. Real-Time Predictions - Due to pre-loading the trained model and pre-processing results into memory at the start of the application, the prediction will be generated quickly enough to enable real-time usability in clinical practice during preparation and execution of the CT scans.

3. **Enhanced Consistency and Decreased Operator Dependency** - Traditional bolus tracking highly depends on the manual setting of the threshold and radiographer's experience. The framework provides standardized predictions based on data, thus enhancing the consistency of the procedure.
4. **Personalized Predictions** - Physiological and protocol parameters (BMI, contrast volume, flow rate) will be considered in order to provide more accurate personalized predictions.
5. **Simple Deployment** - The system uses Streamlit library and locally stored model artifacts. There is no need for complicated cloud infrastructure and data storage which makes the deployment process easy and applicable for hospital workstations or air-gapped clinical environments.
6. **Data Privacy and Security** - Stateless inference is performed by the framework, which does not store or transfer patient information outside the system. This reduces possible risks related to the violation of patient confidentiality.
7. **Scalability and Further Extensibility** - The modular nature of the framework allows adding extra parameters, machine learning algorithms, or even integrating the system with PACS and RIS of the hospital.
8. **Minimization of Radiation Exposure** - With enhanced scan timing accuracy, the possibility of repeat CT scanning due to inadequate contrast enhancement will be reduced, thus minimizing radiation exposure.

CHAPTER 4

METHODOLOGY

4.1 Workflow Description

The current chapter describes in detail the complete machine learning pipeline designed for predicting the CT contrast bolus timing assisted by AI. The pipeline includes six sequential steps: data collection and preprocessing, exploratory data analysis, feature engineering, model training and evaluation, hyperparameter tuning, and finally deploying the model. All the above-mentioned steps are discussed in further sections.

As opposed to existing traditional techniques that use repeated monitoring scans and constant real-time monitoring by the operator during the procedure, the novel technique involves predicting the contrast bolus timing based solely on parameters that were known prior to performing the CT scan itself. This eliminates the necessity for extra monitoring scans in standard cases and reduces operator dependency in scan timing decisions.

4.2 Dataset Description

The dataset used in this study consists of 1,120 patient records collected from real-world CT imaging workflows across clinical settings. After removing duplicate and incomplete entries, 1,108 valid records were retained for analysis. Each record contains both patient-specific and scan-specific attributes collected prior to the CT examination, making the full feature set available before the scan begins.

The input features present in each record are:

- Age (years)
- Height (cm)
- Weight (kg)
- Contrast volume (ml)

- Injection flow rate (ml/s)
- Sex (categorical)
- Contrast agent type (categorical)

The target variable is the bolus tracking time in seconds, representing the time elapsed from the start of contrast injection to the moment the aortic ROI attenuation crosses the predefined HU trigger threshold. The distribution of the target variable has an average value of around 19 seconds and shows significant variance, which can be attributed to the varying physiology found among patients.

4.3 Data Preprocessing

Due to the multi-source and clinical nature of the dataset, a structured preprocessing pipeline was implemented to ensure data consistency, reliability, and suitability for machine learning.

4.3.1 Data Cleaning and Validation

The dataset was examined for missing values, duplicate entries, and data entry errors. Invalid or incomplete records were eliminated, and numeric variables were checked for their data types, ensuring quality data to train the model.

4.3.2 Standardization of Clinical Variables

Variations in the clinical terminology were noted in the contrast agent feature as a result of the lack of uniformity in the way the terms had been recorded in the various source institutions. Terms like “Iopromide 370 mgI” and “Iohexol 350 mgI,” among others, were seen together with abbreviations. The contrast agent terms were categorized into two types using the following classification system:

- Variants containing “Iopromide” or “Ultravist” → Iopromide

- Variants containing “Iohexol” or “Omnipaque” → Iohexol

This standardization ensured semantic consistency across the dataset and prevented fragmentation of the categorical feature during one-hot encoding.

4.3.3 Encoding of Categorical Features

The categorical variables including Sex and contrast agent type were transformed into numeric form using the one-hot encoding technique, whereby the first category was removed to avoid multicollinearity. As a result, two binary indicator variables were produced as Sex_Male and contrast_type_Iopromide.

4.3.4 Feature scaling

Standardization of all the continuous features was performed using Scikit-learn's StandardScaler, where each feature was scaled to have a mean of zero and a standard deviation of one. This scaling was only performed on the training subset while applying it to the testing subset to avoid any form of data leakage. Feature scaling helps ensure that all features contribute equally during model development.

4.3.5 Train-Test Split and Data Leaking Prevention

The dataset was divided into 80% training (886 records) and 20% testing (222 records) sets using a fixed random seed (random_state = 42) for reproducibility. All data preprocessing techniques such as scaling and feature selection were done using parameters obtained only from the training dataset. The test dataset was only used in the final phase of independent evaluation.

4.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed on the processed data set consisting of 1,108 observations. This included the exploration of the distribution of features, the relationship between variables, and pattern recognition in terms of bolus tracking time. EDA covered demographic distribution, target variable assessment, subgroup comparison, correlation analysis, and scatter plots.

4.4.1 Patient Demographics

The patient demographic analysis included age, sex, and contrast agent distributions. Figure 4.1 shows the age distribution for the patients, who are mostly within the age group of 40 to 70 years. The database includes patients from a wide range of ages, indicating diverse cases.

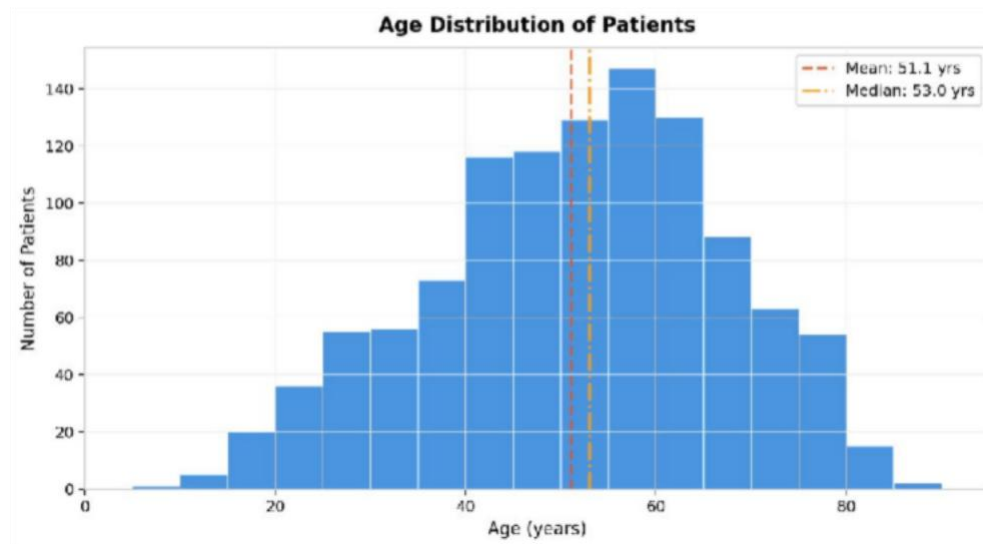


Fig. 4.1 Age distribution of patients in the dataset (n = 1,108)

Figure 4.2 presents the categorical feature distributions for sex and contrast agent type. The dataset contains a higher proportion of male patients compared to female patients. Among contrast agents, Iopromide was used more frequently than Iohexol in the collected dataset.

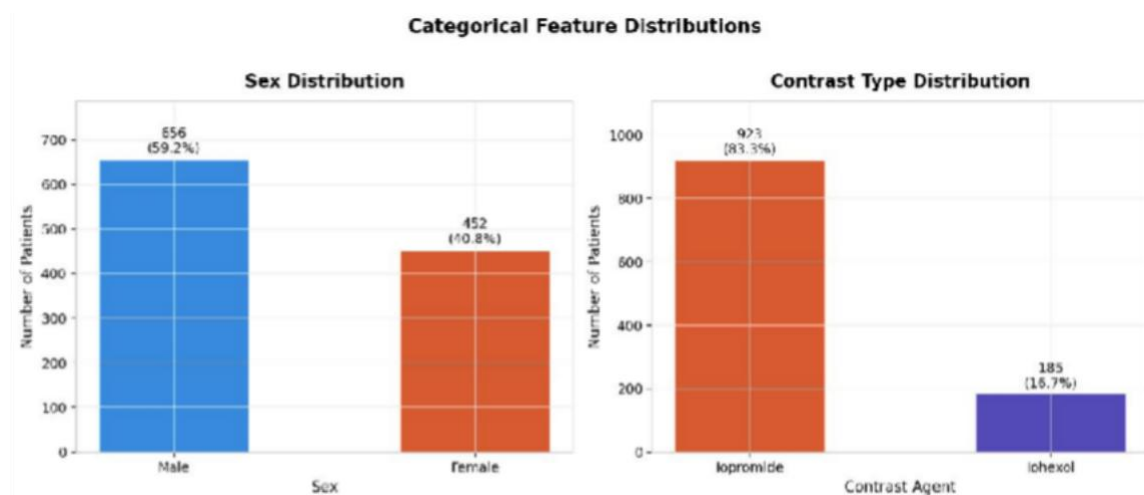


Fig. 4.2 Categorical feature distributions: sex (left) and contrast agent type (right)

4.4.2 Target Variable Distribution

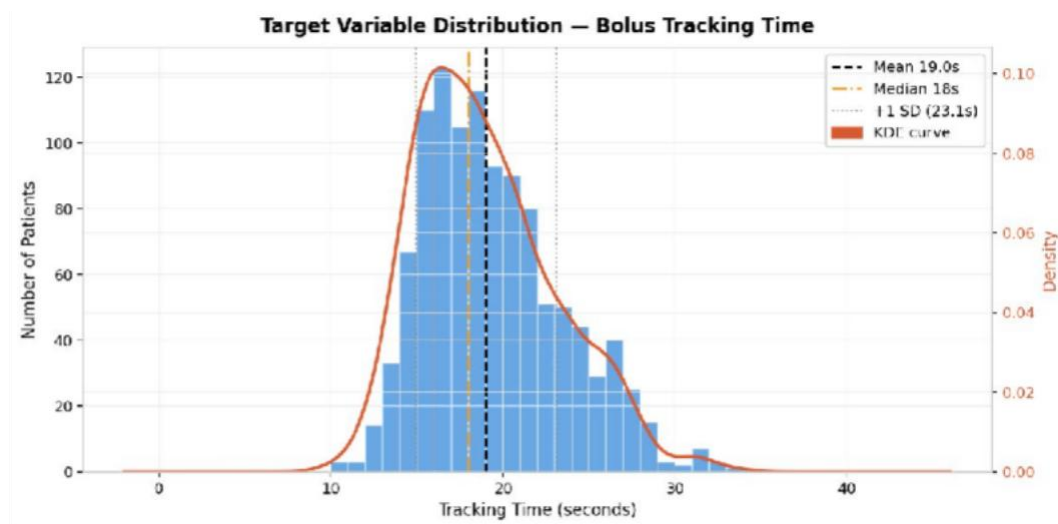


Fig. 4.3 Distribution of bolus tracking time (target variable) across the dataset

Figure 4.3 displays the distribution of the target variable, the bolus tracking time in seconds. The figure shows how the data is distributed, with the majority of values ranging between moderate values and some outliers with higher values.

The analysis of the target variable was very helpful in determining the overall behavior of the dataset and aided in selecting appropriate machine learning models for predicting.

4.4.3 Group Analysis by Patient Subgroups

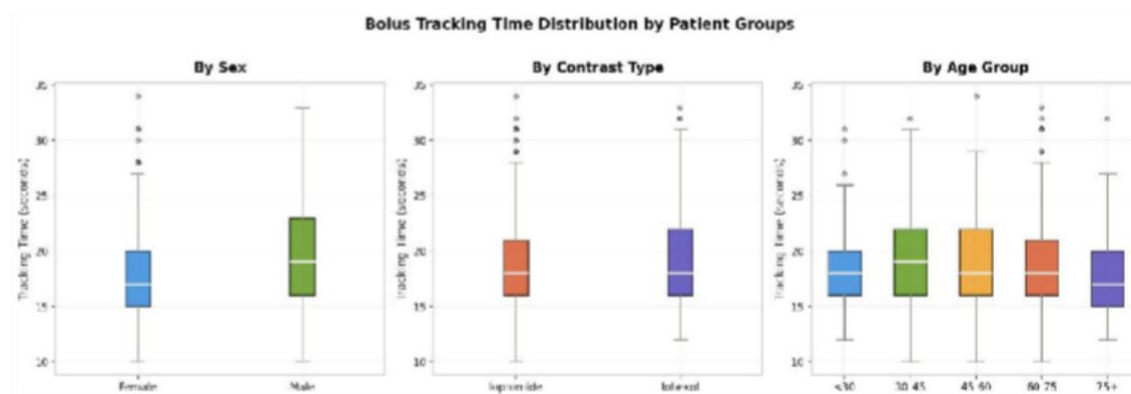


Fig. 4.4 Bolus tracking time distribution by patient sex, contrast agent type, and age group

Bolus tracking time box plots according to patient gender, type of contrast agent used, and age categories are presented in Figure 4.4 below. The box plots were used to analyze the variance in bolus tracking times between various groups.

There are small differences in tracking times between the various groups.

4.4.4 Correlation Analysis

A Pearson correlation heatmap of all numerical features is shown in Figure 4.5. The heatmap was used to identify relationships between features and observe how strongly variables are correlated with one another.

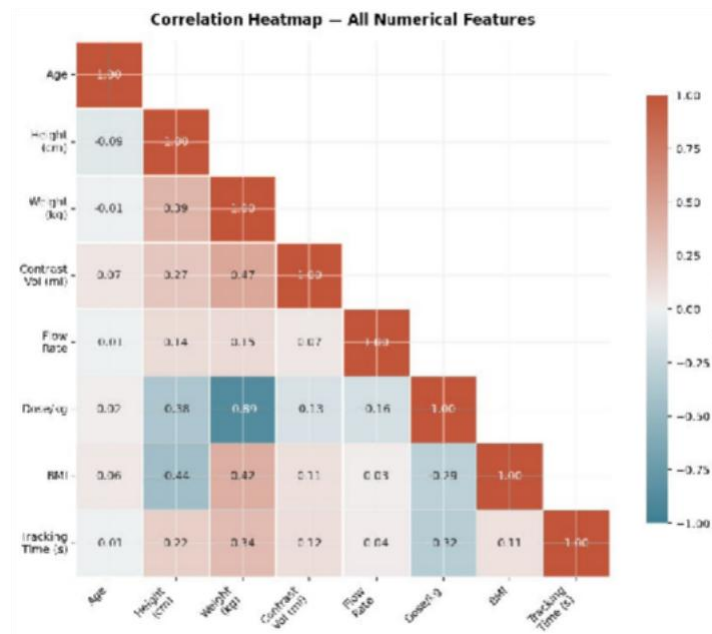


Fig. 4.5 Pearson correlation heatmap of all numerical features

The analysis revealed stronger correlations between some engineered and original features, particularly between dose per kilogram and weight, as well as height and BMI. These observations helped in understanding feature relationships during model development.

4.4.5 Feature-Target Scatter Plots

Important feature versus bolus tracking time scatter plots have been provided in Figure 4.6. Scatter plots have been used to visualize the relationship between each of the selected features and the target feature.

From the scatter plots, it can be seen that there is a different level of relationship between the selected features and bolus tracking time.

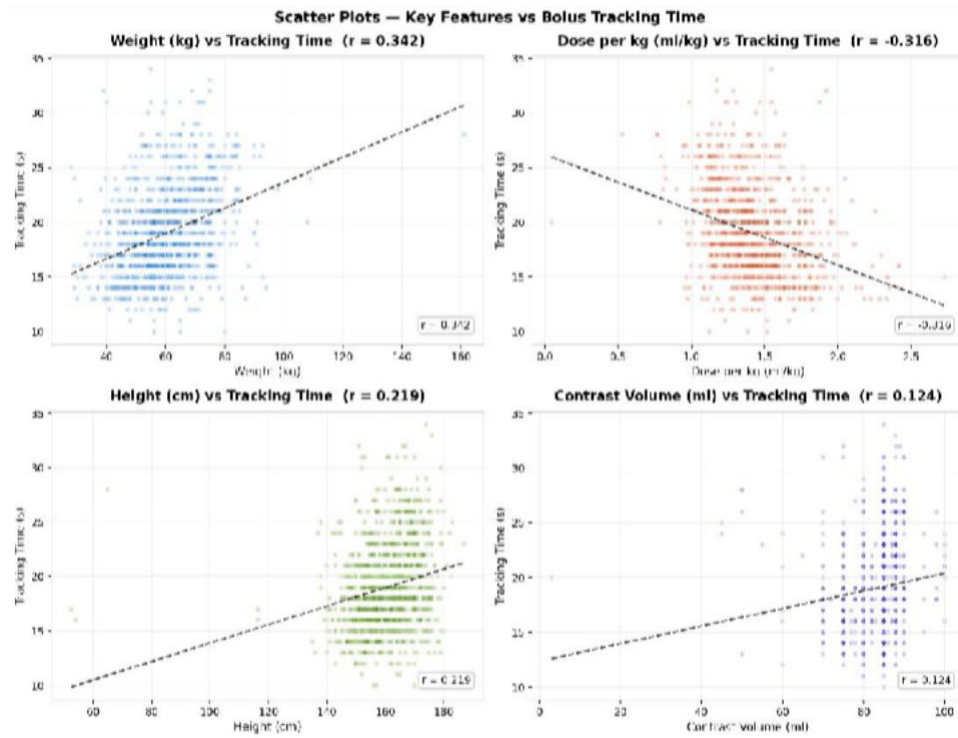


Fig. 4.6 Scatter plots of key features versus bolus tracking time with linear trend lines

4.5 Feature Engineering

Feature engineering was performed to enhance model performance by incorporating clinically meaningful transformations grounded in radiological and physiological principles. Two additional features were derived from the existing raw attributes.

Body Mass Index (BMI) is calculated as weight in kilograms divided by the square of height in meters:

$$\text{BMI} = \text{weight (kg)} / (\text{height (m)})^2 \quad (4.1)$$

Dose per Kilogram is calculated as contrast volume divided by patient body weight:

$$\text{dose_per_kg} = \text{contrast volume (ml)} / \text{weight (kg)} \quad (4.2)$$

The BMI acts as a body habitus factor and affects the blood volume and distribution of contrast within the cardiovascular system. Dose per kilogram reflects the effective contrast concentration relative to patient size and captures the pharmacokinetic relationship between administered dose and body mass. These transformations align with established clinical findings; Shashi Kumar et al. [2] reported significant positive correlations between patient weight and aortic enhancement time ($r = 0.539$, $p < 0.001$), directly supporting the

inclusion of weight-derived features in the predictive model. By explicitly encoding these physiologically meaningful relationships, the model is better equipped to learn complex interactions among input variables.

4.6 Feature Selection

RFE was used with a random forest classifier of 50 trees to evaluate the importance of each feature and find the best set of features. RFE was only trained on the training data to avoid data leakage. The process of removal of the least important features continued until the desired number of features was reached.

The RFE algorithm chose six out of the nine features as being the most predictive: age, height, weight, contrast volume, dose per kilogram, and BMI. The three remaining features, which were removed, were flow rate, sex, and contrast agent type. It is important to mention that the flow rate feature was not chosen because the contribution of this feature is already considered in the dose per kilogram feature.

However, experimentation showed that the models trained on the subset of selected features performed slightly worse than the models trained on all the nine features. Therefore, taking into account the higher variability in the data, it was decided to train the final models on all nine engineered features.

4.7 Models Evaluated

Four regression models were trained and benchmarked across the full pipeline. Each model was selected to represent a different class of learning algorithm, enabling a comprehensive comparison of performance characteristics.

4.7.1 Linear Regression

Linear Regression served as the interpretable baseline model, with the target variable modeled as a linear combination of the features. This model assumes a linear relationship between the predictors and the target and serves as a lower bound for comparison against all non-linear models.

$$\hat{y} = \beta_0 + \sum \beta_i x_i \quad (4.3)$$

4.7.2 Random Forest Regressor

The Random Forest technique builds a forest of T decision trees trained on bootstrapped samples from the training data and combines their outputs through averaging. The process of selecting random subsets of features to perform splits on ensures that the trees are decorrelated, thus decreasing the variance without a significant increase in bias. The model was set up with $T = 100$ estimators.

$$\hat{y}_{\text{RF}} = (1/T) \times \sum \text{fit}(x) \quad (4.4)$$

4.7.3 XGBoost

XGBoost builds trees sequentially, with each tree in the sequence learning from the residual errors made by the previous set of trees in order to minimise a regularised loss function. The regularisation parameter Ω restricts model complexity through penalty on leaf weights and tree depth, thus avoiding overfitting. As a baseline model, the XGBoost model was trained using 100 estimators with a learning rate of 0.1.

$$L = \sum \ell(y_i, \hat{y}_i) + \sum \Omega(f_k) \quad (4.5)$$

4.7.4 Multi-Layer Perceptron (MLP)

The Multi-Layer Perceptron comprised two hidden layers of 100 and 50 neurons, respectively, with Rectified Linear Unit (ReLU) activation functions. The network was trained for up to 1,000 iterations using the Adam optimiser. The MLP was trained on all available features to maximise its opportunity to exploit non-linear interactions in the data.

$$h = \max(0, Wx + b) \quad (4.6)$$

4.8 Hyperparameter Tuning

RandomizedSearchCV with five-fold cross-validation was applied to all three non-linear models to explore whether tuned configurations could improve upon their default baselines. Twenty random combinations were sampled for each model.

Random Forest: The search space covered number of estimators in {100, 200, 300, 400}, maximum depth in {10, 20, 30, None}, minimum samples per split in {2, 5, 10}, minimum samples per leaf in {1, 2, 4}, and maximum features in {sqrt, log2, None}.

XGBoost: The search additionally covered learning rate in {0.01, 0.05, 0.1, 0.2}, maximum depth in {3, 5, 7, 9}, gamma in {0, 0.1, 0.2, 0.3}, subsample in {0.8, 0.9, 1.0}, and colsample_bytree in {0.8, 0.9, 1.0}.

MLP: Architecture configurations including hidden layer sizes, activation functions (ReLU and tanh), regularization strength (alpha), and learning rate schedule were explored across 20 iterations.

4.9 Evaluation Protocol

Mean Absolute Error (MAE) was used as the primary measure throughout the entire pipeline due to its straightforward interpretation in terms of clinical relevance using the unit of seconds. As the secondary measure, Root Mean Square Error (RMSE) was calculated to gauge sensitivity to large errors in predictions.

In relation to the deployed model, a five-fold cross-validation was performed on all 1,108 observations to produce an unbiased estimate of generalization performance irrespective of the initial training and testing data split. This estimate of MAE produced from cross-validation formed the basis for the confidence intervals computed in the deployed application.

4.10 Model Selection and Final Training

After the evaluation of the models using the test dataset and tuning of the hyperparameters, the Random Forest Regressor turned out to be the better-performing model, as it achieved the lowest MAE and RMSE values compared to the other three models. The model was retrained using all the data, i.e., 1,108 observations.

The trained model, the fitted StandardScaler artefact, and the feature column schema were serialised to disk using Joblib as three separate files: rf_model.joblib, scaler.joblib, and model_cols.joblib. These artefacts form the inference core of the deployed web application and are loaded at runtime to ensure that the live prediction pipeline is identical to the

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Baseline Model Performance

To establish a performance reference, all four models were first evaluated on the combined dataset of 1,108 records using raw input features only, with no feature engineering and no hyperparameter tuning. Results are summarized in Table 5.1.

Table 5.1: Baseline Model Performance (No Feature Engineering)

Model	MAE (s)	RMSE (s)
Linear Regression	3.085	3.89
Random Forest	2.772	3.53
XGBoost	2.942	3.75
ANN (MLP)	3.085	5.25

Random Forest achieved the lowest baseline MAE of 2.772 seconds, followed by XGBoost at 2.942 seconds. Both Linear Regression and the ANN recorded a MAE of 3.085 seconds at baseline, with the ANN additionally exhibiting a notably higher RMSE of 5.25 seconds, indicative of convergence instability under the default configuration. The performance gap between the linear baseline and the ensemble methods confirms that non-linear relationships are present in the data that a linear model cannot capture.

5.2 Impact of Feature Engineering

Following the introduction of BMI and dose per kilogram as engineered features, all models were re-evaluated on the full nine-feature set. Results are summarized in Table 5.2.

Table 5.2: Model Performance After Feature Engineering

Model	Baseline MAE (s)	Post-Engineering MAE (s)	Improvement (s)
Linear Regression	3.085	3.085	0.000
Random Forest	2.772	2.717	+0.055
XGBoost	2.942	2.848	+0.094
ANN (MLP)	3.085	2.847	+0.238

Feature engineering improved the performance of Random Forest, XGBoost, and the MLP. The most substantial improvement was observed in the MLP (0.238 seconds), which benefited significantly from the explicit encoding of physiologically meaningful interactions. Random Forest improved by 0.055 seconds and XGBoost by 0.094 seconds. There was no improvement in the case of Linear Regression, as the created features can be mathematically expressed in terms of the raw variables already present in the data. These results validate the importance of domain-informed feature transformations in clinical prediction tasks.

5.3 Feature Selection Results

The six features which were found to be the most important according to the Recursive Feature Elimination technique include: age, height, weight, contrast volume, dose per kg, and BMI. The features which were removed by RFE include: flow rate, gender, and contrast type.

However, the models that were built based on the six features performed slightly worse than the models that were built based on the nine features in terms of the test set performance. The conclusion can be drawn that the features removed by the RFE algorithm have informative predictive power. Thus, the final models were built based on all nine features.

5.4 Final Model Comparison

Table 5.3 presents the final performance of all four models trained on the complete engineered feature set with their respective optimal configurations.

Table 5.3: Final Model Performance Comparison

Model	MAE (s)	RMSE (s)
Linear Regression	3.085	3.89
Random Forest	2.717	3.53
XGBoost	2.837	3.75
MLP (ANN)	3.215	5.25

Random Forest achieved the lowest MAE of 2.717 seconds, confirming it as the best-performing model for deployment. XGBoost followed at 2.837 seconds, while the MLP recorded the highest final error at 3.215 seconds due to persistent convergence issues. Linear Regression remained unchanged at 3.085 seconds, limited by its inability to model the non-linear physiological relationships present in the data.

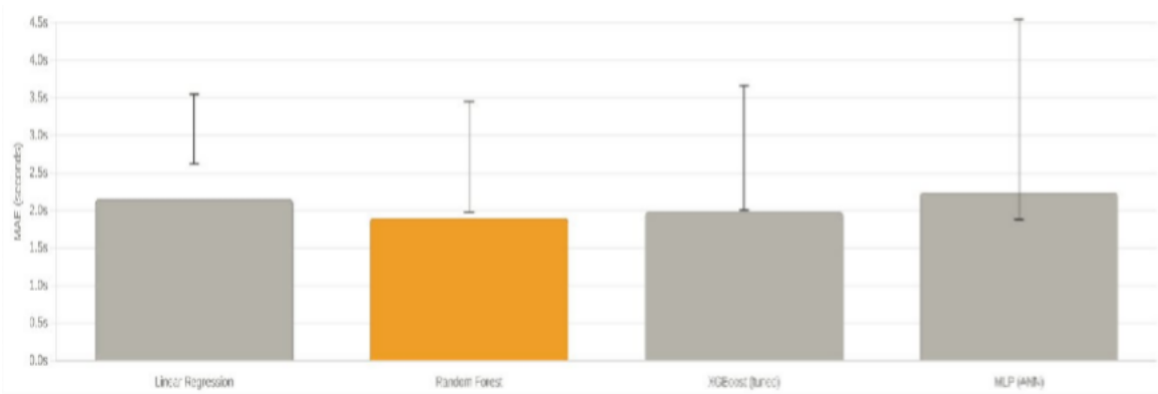


Fig. 5.1: Cross-Validated MAE Comparison Across All Four Regression Models

As illustrated in Fig. 5.1, Random Forest consistently records the lowest MAE with the tightest error bars, further demonstrating its stability and generalization capability across cross-validation folds.

5.5 Hyperparameter Tuning Results

RandomizedSearchCV for five folds of cross-validation was performed on all three models, namely the Random Forest, XGBoost, and MLP models. In the case of Random Forest, the fine-tuning of hyperparameters had little impact on the MAE value, increasing it by only 0.025 seconds, from 2.717 seconds to 2.742 seconds. This means that the default Random Forest configuration generalizes better than the tuned configuration to the given dataset size. Therefore, the non-fine-tuned Random Forest is selected as the final model to avoid overfitting to the cross-validation folds. For XGBoost, the best performing combination of parameters included `n_estimators = 200`, `max_depth = 7`, `learning_rate = 0.1`, `gamma = 0.3`, `subsample = 0.9`, `colsample_bytree = 0.8`, achieving the MAE of 2.837 seconds. The MLP showed improvements due to the architectural tuning; however, it still performed worse compared to the other two non-linear models.

5.6 Feature Importance Analysis

Feature importance was extracted from the final Random Forest model using the mean decrease in impurity criterion. The ranked importances are as follows:

Table 5.4: Random Forest Feature Importance Rankings

Feature	Importance (%)
Weight (kg)	18.4
BMI	18.3
Age	17.7
Dose per kg	13.7
Height (cm)	13.1
Contrast Volume (ml)	7.8

Flow rate(ml/s)	6.1
Sex	2.9
Contrast type	2.0

As shown in Fig. 5.2, patient weight, BMI, and age collectively account for over 54% of predictive importance, establishing body composition and patient age as the dominant drivers of bolus tracking time. This finding aligns with the clinical observations of Shashi Kumar et al. [2], who reported a significant positive correlation between patient weight and aortic enhancement time ($r = 0.539$, $p < 0.001$). The two engineered features, BMI and dose per kilogram, together contribute approximately 32% of total importance, confirming that feature engineering meaningfully expanded the model's predictive capacity beyond what the raw inputs alone could provide. Flow rate ranked lowest among retained features at 6.1%, consistent with the RFE finding that it was excluded from the optimal six-feature subset.

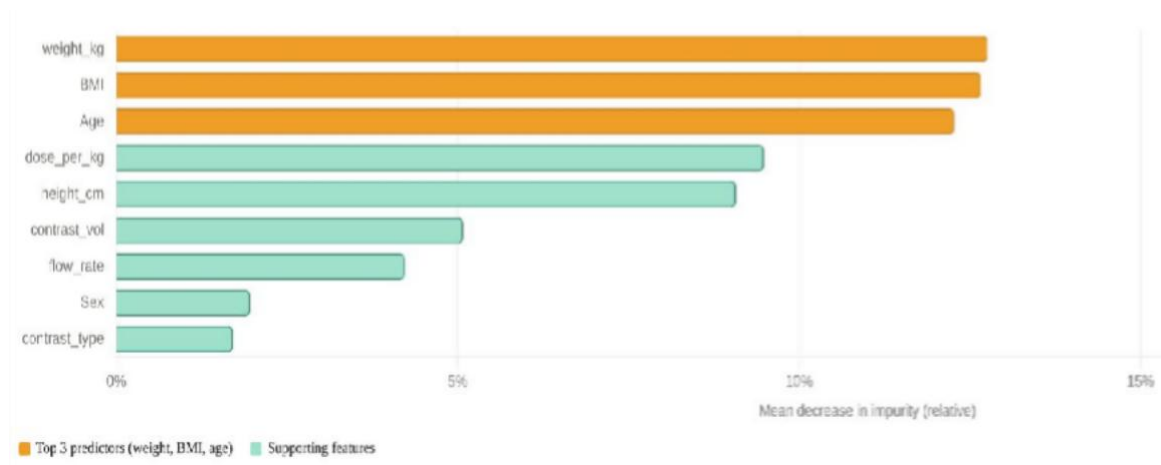


Fig. 5.2: Random Forest Feature Importance Rankings (Mean Decrease in Impurity)

5.1 Final Cross-Validated Performance

To ensure an unbiased final assessment of the performance, a five-fold cross-validation was performed on the entire dataset containing 1,108 rows using the final Random Forest model. This method helps prevent any optimistic reporting of errors due to a favorable train-test split.

The cross-validated MAE of 2.693 seconds (± 0.737 seconds) is marginally lower than the train-test MAE of 2.717 seconds, indicating that the model generalizes consistently across different subsets of the data and is not dependent on a particular train-test partition.

5.2 Validation on Unseen Clinical Data

To further evaluate real-world predictive performance, the deployed model was tested on four unseen patient records not included in the training or test sets. Results are presented in Table 5.5.

Table 5.5: Prediction Results on Unseen Clinical Records

Actual Time (s)	Predicted Time (s)	Lower Bound (s)	Upper Bound (s)	Absolute Error (s)
26	17.6	14.9	20.3	8.4
26	23.0	20.3	25.7	3.0
18	18.7	16.1	21.4	0.7
20	18.8	16.1	21.5	1.2

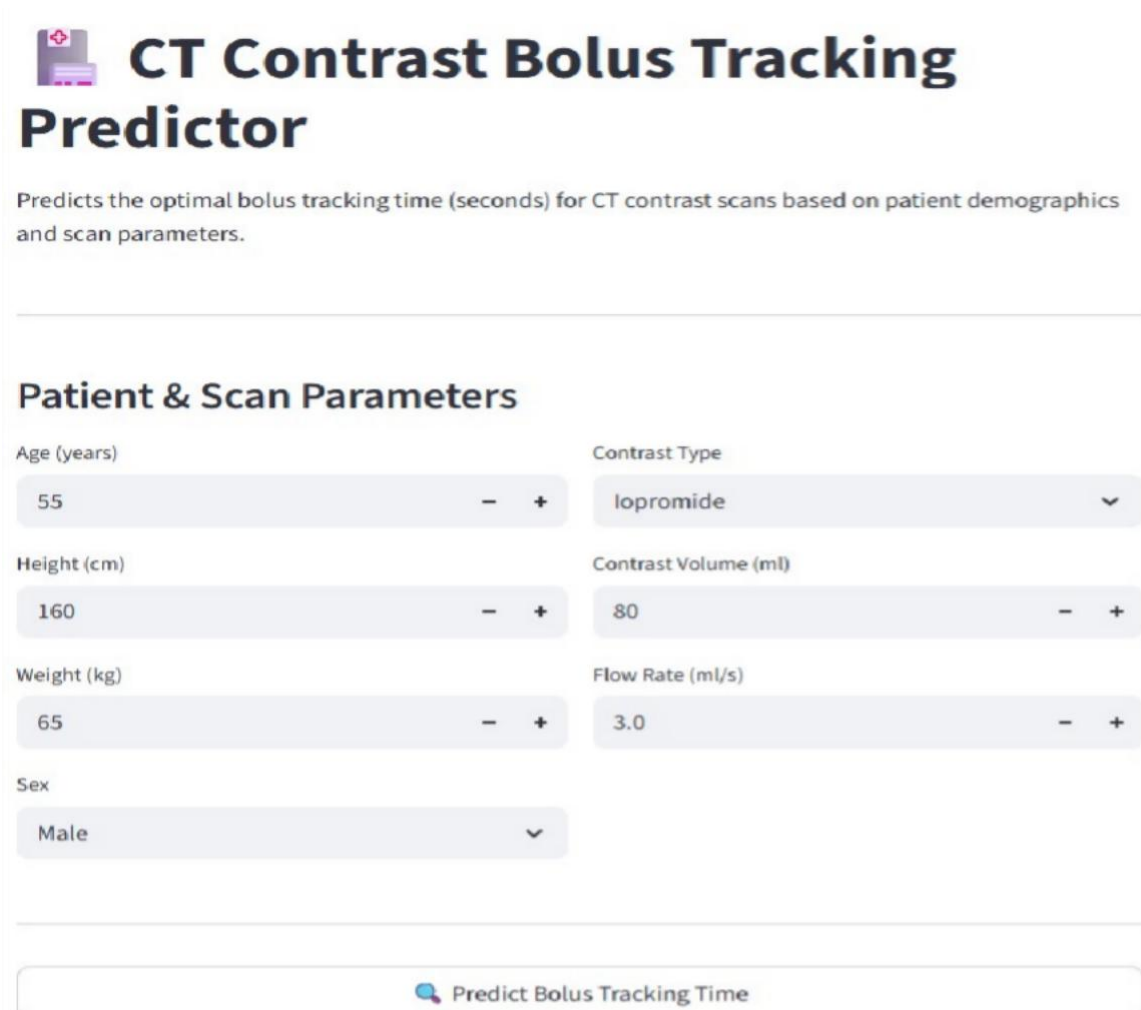
Three of the four unseen predictions lie within 3 seconds of the true recorded bolus-tracking time, with two of the predictions lying within 1.5 seconds of the true values. The only one that lies outside this range is the prediction of 17.6 seconds when the true value was 26 seconds (absolute error = 8.4 seconds), which lies on the higher side of the bolus-tracking time distribution, a segment that is not well-represented in the training data set. This result agrees with the known behavior of regression models trained on skewed target distributions, where predictions made in sparsely populated areas of the target space tend to be less reliable. The prediction intervals calculated based on the MAE of ± 2.693 seconds from cross-validation include the true value in three out of four cases.

These results confirm the model's practical viability for typical clinical scenarios while highlighting that further data collection targeting patients with longer bolus tracking times would improve accuracy in edge cases.

5.3 Streamlit Web Application

The final Random Forest model, trained on the complete 1,108-record dataset, was deployed as a Streamlit web application. The interface accepts seven raw clinical parameters, age, height, weight, sex, contrast type, contrast volume, and flow rate and automatically computes BMI and dose per kilogram. The scaled feature vector is passed to the saved model, and the application returns the predicted bolus tracking time alongside a lower and upper uncertainty bound of ± 2.693 seconds.

Figs. 5.3 and 5.4 show the input form and the prediction output screen respectively.



The screenshot displays the 'CT Contrast Bolus Tracking Predictor' web application. At the top, there is a title 'CT Contrast Bolus Tracking Predictor' with a sub-description: 'Predicts the optimal bolus tracking time (seconds) for CT contrast scans based on patient demographics and scan parameters.' Below this, the 'Patient & Scan Parameters' section contains seven input fields arranged in two columns. The left column includes 'Age (years)' (55), 'Height (cm)' (160), 'Weight (kg)' (65), and 'Sex' (Male). The right column includes 'Contrast Type' (Iopromide), 'Contrast Volume (ml)' (80), and 'Flow Rate (ml/s)' (3.0). Each numerical field has minus and plus buttons for adjustment. At the bottom, there is a large 'Predict Bolus Tracking Time' button.

Fig. 5.3: Streamlit Application (Patient Input Form)

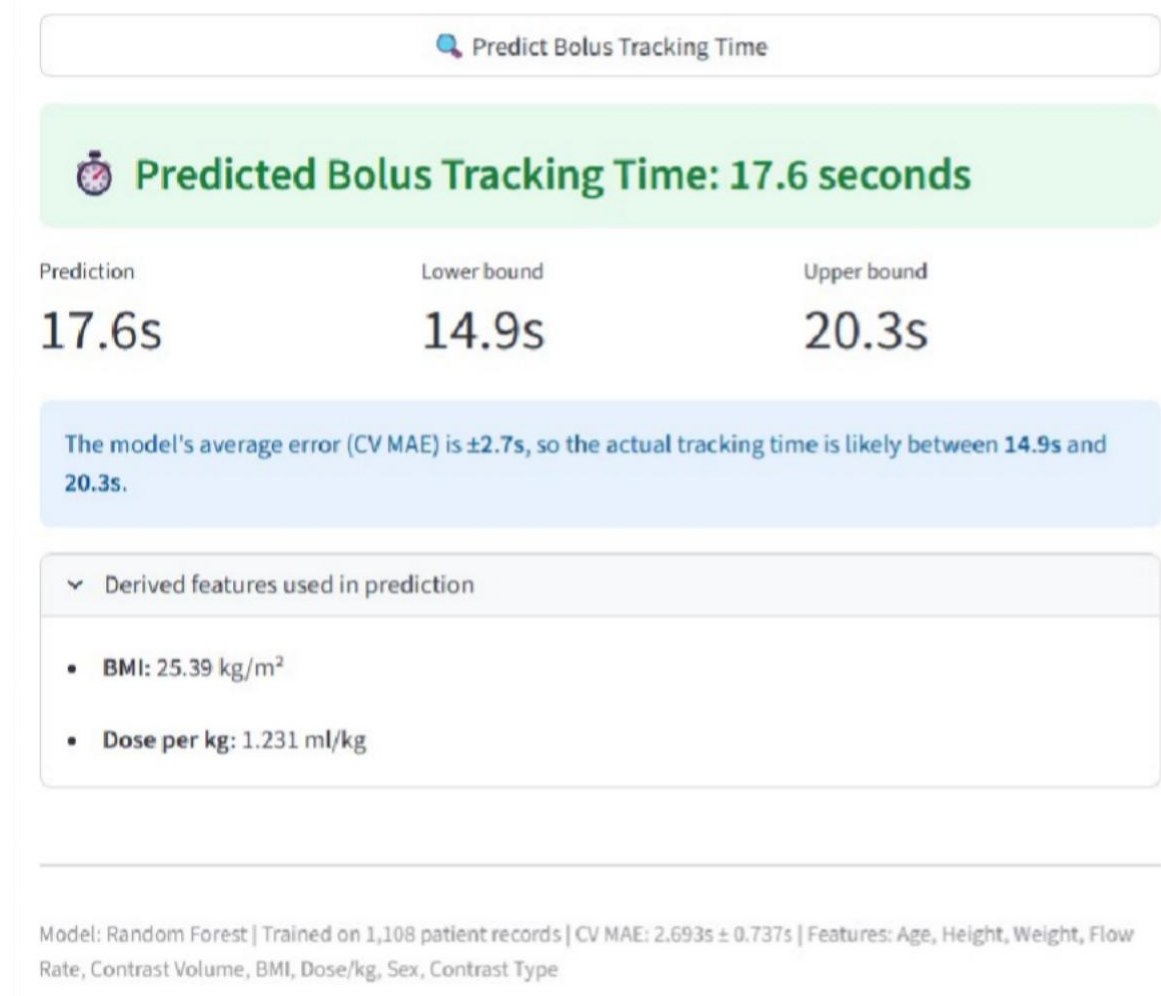


Fig. 5.4: Streamlit Application (Prediction Output with Confidence Bounds)

The application is designed as a pre-scan decision-support tool. A radiographer can review the predicted tracking time before contrast injection and use it to plan scan timing, reducing the need for repeated low-dose monitoring acquisitions in typical clinical cases.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

Accurate estimation of bolus tracking time is crucial for obtaining high-quality contrast-enhanced CT images. Conventional methods of estimating bolus tracking time with predefined HU thresholds fail to consider patient-specific variability. Therefore, in this study, a machine learning model was created to predict bolus tracking time using routine clinical data before the contrast medium administration.

For this purpose, a dataset containing information of 1,108 patients was used in addition to two engineered features – BMI and dose per kilogram. Four regression models were trained and compared. Among them, the Random Forest Regressor showed the best results with a test MAE of 2.717 seconds and cross-validated MAE of 2.693 ± 0.737 seconds. Hyperparameters tuning indicated that the default configuration of the Random Forest generalizes better than the tuned version. This means that the Random Forest Regressor is suitable for the current dataset.

The analysis of feature importances showed that weight, BMI, and age made up 54% of the model's performance. It corresponds to clinical knowledge because weight and BMI affect the speed of blood flow and thus the time of contrast transit through tissues. The engineered features – BMI and dose per kilogram – accounted for 32% of total feature importances, proving the advantage of domain-based feature engineering.

Finally, a Streamlit web application that takes seven input features and makes predictions with an expected uncertainty of ± 2.693 seconds was deployed. Four unknown patient records were used for testing the app. The predictions were found to be accurate with differences less than 3 seconds for three out of four patients. Thus, the model can be used as a decision-making tool prior to the scan. The fourth patient had a predicted bolus tracking time of 26 seconds, which was underestimated by the model due to lack of such cases in the dataset.

In conclusion, this study shows that a machine learning algorithm can provide reasonable prediction accuracy when predicting bolus tracking time in CT scanning.

6.2 Future Work

The results of this project highlight several areas for future improvement and development.

6.2.1 Dataset Expansion

The first and foremost direction for improvement is increasing the size of the dataset, particularly for patients requiring longer bolus tracking. The current dataset has a smaller number of examples of patients that require 24-30 seconds to complete the contrast medium passage. An increase in the amount of examples of this type will benefit the learning process and the accuracy of predicting atypical patients' needs.

It would also be beneficial to collect additional data about patients with cardiac dysfunction, high BMI values, and other physiological characteristics that might affect contrast transit.

6.2.2 Mobile Application Development

The current solution is implemented as a web application using the Streamlit framework. Although this implementation works, it is not entirely convenient and can be improved.

Creating an Android/iOS app would significantly enhance user experience and simplify the process of obtaining results since predictions can be made offline without access to a browser and a server.

6.2.3 External Validation and Multi-Centre Evaluation

The training of the model took place using data from one particular institution. To test the ability of the model to generalize across other patients, it would be essential to conduct external validation on datasets obtained from several hospitals and scanners.

REFERENCES

- [1] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [2] Shashi Kumar, S. Sukumar, W. Dkhar, and S. Babu, "Evaluation of factors affecting intravenous CT contrast medium kinetics," *Int. J. Current Research*, vol. 7, no. 7, pp. 17906–17911, Jul. 2015.
- [3] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, 2016, pp. 785–794.
- [4] C. T. Jensen, R. Khetan, J. Adkins, S. Javadi, X. Liu, J. Sun, S. A. Hassan, and A. C. Morani, "Delayed bolus-tracking trigger at CT correlates with cardiac dysfunction and suboptimal portovenous contrast phase," *Abdom. Radiol.*, vol. 46, no. 2, pp. 826–835, Feb. 2021, doi: 10.1007/s00261-020-02655-8.
<https://pubmed.ncbi.nlm.nih.gov/32700213/>
- [5] Y. Noda, N. Kawai, T. Ishihara, Y. Tsuboi, T. Kaga, T. Miyoshi, F. Hyodo, and M. Matsuo, "Optimized scan delay for late hepatic arterial or pancreatic parenchymal phase in dynamic contrast-enhanced computed tomography with bolus-tracking method," *Br. J. Radiol.*, vol. 94, no. 1122, Jun. 2021, Art. no. 20210315, doi: 10.1259/bjr.20210315.
<https://pubmed.ncbi.nlm.nih.gov/33956484/>
- [6] J. Yu, S. Lin, H. Lu, R. Wang, J. Liu, R. Gutjahr, and J. Gao, "Optimize scan timing in abdominal multiphase CT: Bolus tracking with an individualized post-trigger delay," *Eur. J. Radiol.*, vol. 148, Art. no. 110139, Mar. 2022, doi: 10.1016/j.ejrad.2021.110139.
<https://www.sciencedirect.com/science/article/abs/pii/S0720048X21006203>
- [7] I. Heukensfeldt Jansen, E. Haneda, B. Claus, P. Wu, J. D. Pack, and B. De Man, "Estimation of contrast agent concentration from pulsed-mode projections to time contrast-enhanced CT scans," in *Proc. 7th Int. Conf. Image Formation X-*

- Ray Comput. Tomogr.*, 2022, vol. 12304, Art. no. 123041Z, doi: 10.1117/12.2647156.
<https://www.spiedigitallibrary.org/conference-proceedings-of-spie/12304/123041Z/Estimation-of-contrast-agent-concentration-from-pulsed-mode-projections-to/10.1117/12.2647156.full>
- [8] A. T. Li, P. B. Noël, and N. Shapira, "Automatic bolus tracking in abdominal CT scans with convolutional neural networks," *Quant. Imaging Med. Surg.*, vol. 13, no. 5, pp. 2780–2790, May 2023, doi: 10.21037/qims-22-686.
<https://pubmed.ncbi.nlm.nih.gov/37179937/>
- [9] J. H. Song, D. Park, Y. S. Jeong, and J. K. Park, "Potential role of bolus-tracking data of carotid CT angiography for atrial fibrillation prediction," *Eur. Radiol.*, vol. 33, no. 2, pp. 981–987, Feb. 2023, doi: 10.1007/s00330-022-09072-z.
<https://pubmed.ncbi.nlm.nih.gov/35962815/>
- [10] F. Escher, N. Fink, S. Maurus, J. Dinkel, J. Ricke, J. Hausleiter, and A. Curta, "Optimization of Preprocedural Full-cycle Computed Tomography in Patients Referred for Transcatheter Tricuspid Valve Repair: Test Bolus Versus Bolus Tracking," *J. Thorac. Imaging*, vol. 38, no. 3, pp. 174–178, May 2023, doi: 10.1097/RTI.0000000000000682.
<https://pubmed.ncbi.nlm.nih.gov/33956484/>
- [11] P. T. Boonen, N. Buls, G. van Gompel, H. Devos, Y. de Brucker, T. Leiner, D. Aerden, J. de Mey, and J. Vandemeulebroucke, "Quantitative hemodynamic assessment of stenotic below-the-knee arteries using spatio-temporal bolus tracking on 4D-CT angiography," *Med. Phys.*, vol. 50, no. 11, pp. 6844–6856, Nov. 2023, doi: 10.1002/mp.16755.
<https://pubmed.ncbi.nlm.nih.gov/37750537/>
- [12] M. D. Yuan, Y. Zhang, J. Li, X. Huang, L. Wang, and T. Chen, "Comparative Study of Artificial Intelligence Triggering Technique and Conventional Bolus Tracking Triggering Technique in Coronary CT Angiography," *J. Med. Imaging Radiat. Sci.*, vol. 55, no. 1, pp. 123–130, Mar. 2024.
<https://www.sciencedirect.com/science/article/abs/pii/S1939865424003059>

- [13] S. Watanabe, K. Sakaguchi, D. Murata, and K. Ishii, "Deep learning-based Hounsfield unit value measurement method for bolus tracking images in cerebral computed tomography angiography," *Radiol. Phys. Technol.*, vol. 17, no. 1, pp. 123–132, Mar. 2024, doi: 10.1007/s12194-023-00762-w.
<https://www.sciencedirect.com/science/article/abs/pii/S0010482521006181>
- [14] H. Wu, J. Wang, M. Zhou, Y. Wang, C. Cui, C. Zhou, X. Chen, and Z. Wang, "Application of bolus tracking: The effect of ROI positions on the images quality of cervicocerebral CT angiography," *Heliyon*, vol. 10, no. 7, Art. no. e29260, Apr. 2024, doi: 10.1016/j.heliyon.2024.e 29260.
<https://pubmed.ncbi.nlm.nih.gov/38617933/>
- [15] A. Murphy and H. White, "A comparison of bolus track and test bolus computed tomography pulmonary angiography and the implications on pulmonary and aortic vessel enhancement, effective dose and suboptimal scan rate," *J. Med. Radiat. Sci.*, vol. 71, no. 1, pp. 44–50, Mar. 2024, doi: 10.1002/jmrs.724.
<https://pubmed.ncbi.nlm.nih.gov/37675768/>
- [16] R. Valletta, M. Bonatti, V. Vingiani, V. Corato, B. Proner, F. Lombardo, G. Avesani, P. Pertner, and G. A. Zamboni, "Feasibility of a single-phase portal venous CT protocol using bolus tracking technique and lean body weight-based contrast media dose," *Eur. Radiol.*, vol. 35, no. 2, pp. 1067–1075, Feb. 2025, doi: 10.1007/s00330-024-11009-7.
<https://pubmed.ncbi.nlm.nih.gov/39122853/>
- [17] K. Qi, L. Li, D. Yuan, Y. Zhang, M. Zhang, W. Zhang, Y. Gu, Z. Wang, P. Lyu, J. Gao, and J. Liu, "Optimized contrast enhancement and homogeneity in aortic CT angiography: bolus tracking with personalized post-trigger delay," *Quant. Imaging Med. Surg.*, vol. 15, no. 1, pp. 709–720, Jan. 2025, doi: 10.21037/qims-24-0624.
<https://pubmed.ncbi.nlm.nih.gov/39839042/>
- [18] L. Nas, B. F. Hoppe, A. T. Stüber, S. Grosu, N. Fink, A. von Fragstein, J. Rudolph, J. Rieke, and B. O. Sabel, "Optimizing lower extremity CT angiography: A prospective study of individualized vs. fixed post-trigger delays in bolus tracking," *Eur. J. Radiol.*, vol. 185, Art. no. 112009, Apr. 2025, doi:

10.1016/j.ejrad.2025.112009.

<https://pubmed.ncbi.nlm.nih.gov/40022840/>

- [19] L. Giansante, E. McDonagh, J. Basso, A. Haris, S. Kc, S. J. Withey, J. Shur, N. Fotiadis, S. N. Goldberg, and E. W. Johnston, "Bolus-Tracked Biphasic Contrast-Enhanced CT Imaging Following Microwave Liver Ablation Improves Ablation Zone Conspicuity and Semi-automatic Segmentation Quality," *Cardiovasc. Intervent. Radiol.*, vol. 48, no. 5, pp. 643–652, May 2025, doi: 10.1007/s00270-024-03948-x.
<https://pubmed.ncbi.nlm.nih.gov/39789263/>
- [20] E. Haneda, I. M. Heukensfeldt Jansen, P. Wu, J. D. Pack, A. Hsiao, E. McVeigh, and B. De Man, "Bolus tracking from pulsed x-ray projections: A feasibility study using a five-dimensional cardiac CT contrast dynamics model," *Med. Phys.*, vol. 52, no. 1, pp. 131–145, Jan. 2025, doi: 10.1002/mp.17464.
<https://pubmed.ncbi.nlm.nih.gov/39413309/>

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